DEMONSTRATION METER RANGE ATTACHMENTS

Operating Instructions for Modular Multipliers & Shunts

PHYSICS LABORATORY GUIDE
Base Instrument Specification:
Full-Scale
Deflection

Overview: These modular black plastic attachment blocks are interchangeable range multipliers (for voltage) and shunts (for current). They attach directly to a standard educational moving-coil or moving-iron base demonstration meter movement to convert it into a multi-range AC/DC Voltmeter or Ammeter.

1. System Principles & Technical Function

The base meter movement inherently responds to a small full-scale input of $10\ mA$ current or $100\ mV$ potential drop across its internal resistance ($R_m = 10\ \Omega$). The attachment blocks adjust these electrical parameters as follows:

Attachment Type	Labels / Examples	Internal Circuitry	Function
DC Voltage Multipliers	1V DC, 2.5V DC, 100V DC	Precision series resistor (R_s)	Drops excess voltage so that full scale voltage produces exactly $10\ mA$ through the base meter.
AC Voltage Multipliers	20V AC, 300V AC	Series resistor (R_s) + Rectifier bridge	Rectifies AC to DC and drops excess voltage to match base instrument deflection.
DC Current Shunts	50 mA, 1A DC, 5A DC	Low-resistance parallel shunt (R_{sh})	Bypasses majority of circuit current, maintaining exactly $100\ mV$ ($10\ mA$) across the meter at full scale.

2. Step-by-Step Operating Instructions

1. **Select the Appropriate Range Attachment:** Identify the parameter to be measured (AC/DC, Voltage or Current) and choose a block with a full-scale range exceeding the maximum expected measurement.
2. **Attach Block to Base Meter:** Plug the two terminal pins on the back/top of the block firmly into the receiving banana sockets on the base demonstration meter instrument. Ensure tight mechanical and electrical contact.
3. **Wire into Circuit:**
   - **Voltage Blocks (Voltmeters):** Connect the circuit leads across (in *parallel* with) the component whose potential difference is being measured.
   - **Current Blocks (Ammeters):** Connect the circuit leads in *series* with the load line so that circuit current flows through the block terminals.
4. **Read the Measurement:** Read the pointer deflection on the base meter scale and apply the appropriate scale factor matching the attached block's designated full-scale value.

3. Internal Resistance Calculations (For Physics Lab Exercises)

For educational demonstrations involving internal meter resistance calculations, use the following standard equations:

Voltmeter Series Resistance Calculation:

To measure voltage V using base movement ($I_m = 10\ mA$, $V_m = 100\ mV$):

$$R_s = (V / I_m) - R_m = (V / 0.010\ A) - 10\ \Omega$$

Example for 100V DC block: $R_s = (100 / 0.010) - 10 = 9{,}990\ \Omega$.

Ammeter Shunt Resistance Calculation:

To measure current I using base movement ($V_m = 0.1\ V$):

$$R_{sh} = V_m / (I - I_m) = 0.1\ V / (I - 0.010\ A)$$

Example for 5A DC block: $R_{sh} = 0.1 / (5 - 0.010) = 0.1 / 4.990 \approx 0.02004\ \Omega$.

SAFETY PRECAUTIONS:

- Never connect a Current Shunt block in parallel across a voltage source; doing so will cause a short circuit and damage the shunt resistor.
- Always start circuit testing with the highest voltage or current range block installed if the signal magnitude is unknown.
- Observe correct polarity (+ / red to positive) for DC ranges to prevent reverse needle pegging.
- Do not exceed high voltage ratings (e.g., 300V AC) without verifying isolation integrity.